Vishal V S

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github.com/vishalvijay-123 vishalvs-portfolio.vercel.app Kerala

Profile

Software Engineer with hands-on experience designing and building full-stack web applications, REST APIs, AI-powered automation tools, and embedded IoT systems. Proficient in JavaScript, Python, Java, React, Next.js, Node.js, FastAPI, Docker, Linux, and SQL. Passionate about building scalable software, solving complex engineering problems, and learning modern backend technologies.

Work Experience

Embedded Systems & IoT Intern, *HCL Technologies* 01/2026 – 04/2026

- Designed and built a Smart Hydroponics IoT automation system using ESP32 microcontrollers to monitor temperature, humidity, and water levels, automating relay-controlled irrigation pumps based on real-time sensor thresholds.
- Programmed C++ firmware for ESP32 with multi-sensor integrations (DHT22, water-level sensors), relay module control, and Wi-Fi connectivity, achieving 99.9% uptime in testing.
- Configured and maintained Linux development environments for embedded firmware development, backend services, and application deployment.

Stack: ESP32, C++, Arduino IDE, FastAPI, SQLite, Python

Projects

VS. — Full-Stack Company Website

- Built a full employee management web app (Next.js, React, TypeScript) with separate dashboards for Admins, Managers, and Employees — including task boards, leave requests, attendance tracking, and live notifications.
- Designed the database structure (10 linked tables) and built 30+ secure APIs with role-based access control.
- Created an AI tool that reads resume PDFs and automatically scores candidates, cutting manual screening time by 90%.
- Added secure login with password hashing, session cookies, and safe file uploads.

Stack: Next.js, React, TypeScript, Prisma, SQLite, Tailwind CSS, Node.js

https://github.com/vishalvijay-123/FullStackCompanywebsite

Old Skool Boxing Gym — Client Website & CMS

- Built and deployed a full-stack client website with an animated frontend and a 5-tab CMS admin panel, running live on Render with 24/7 uptime.
- Optimized image uploads using in-memory Base64 compression, avoiding disk writes and reducing upload times.
- Set up a flexible database system (SQLite for development, PostgreSQL/Supabase for production) with automatic table setup, default data seeding, and a scheduled job to clean up old messages every 10 days.

Stack: Vite, Node.js, Express, PostgreSQL, Supabase, JWT, Render

https://github.com/vishalvijay-123/oldskoolboxing
https://oldskoolboxing.onrender.com/

Web Application Penetration Testing

- Conducted black-box penetration testing on 5 live web applications, identifying critical vulnerabilities including SQL Injection, Cross-Site Scripting (XSS), and CORS misconfigurations.
- Authored detailed remediation reports with prioritized findings and proof-of-concept walkthroughs for 100% of target owners.

Stack: Burp Suite, Nmap, Kali Linux, SQLmap, Dirbuster

Custom Vulnerability CTF Box
- Designed and hosted a custom boot-to-root Linux challenge machine on VirtualBox containing multiple exploit chains (privilege escalation, directory traversal) to train peer security analysts.

Stack: VirtualBox, Kali Linux, Metasploit, Burp Suite

Education

B.E. Electronics and Communication Engineering, *Aarupadai Veedu Institute of Technology, Chennai*
CGPA: 8.4

Advanced Diploma in Cyber Defense, *Red Team Hacker Academy, Thiruvananthapuram*
Score: 86%

12th Standard, *Christ Nagar Senior Secondary School, Thiruvananthapuram*
Score: 86.6%

10th Standard, *Amrita Vidyalayam*
Score: 100%

Technical Skills

Frontend
React 19, Next.js 16 (App Router, RSC), TypeScript, Tailwind CSS, shadcn/ui, Radix UI, TanStack Query v5, Zustand

Databases
PostgreSQL, SQLite, MySQL, MongoDB, Supabase

AI & Automation
Ollama, DeepSeek, SDXL, Local LLM Deployment, Deterministic Scoring Engines, pdf-parse

Linux & Infrastructure
Linux Administration (Fedora, Ubuntu, Arch, Kali), Docker & Docker Compose, Git, VirtualBox, Bash, Networking Fundamentals (TCP/IP, DNS, Firewalls)

Backend & APIs
Node.js, Express, FastAPI (Python), REST API Design, Prisma ORM, bcrypt, JWT, cookie-based auth

Cybersecurity
Burp Suite, Metasploit, Nmap, Wireshark, Kali Linux, RBAC, SIEM (ELK, Splunk), CEH Certified

Embedded & IoT
ESP32, Arduino, ATtiny85, C/C++, ESP-NOW, Wi-Fi, GPIO, UART, I2C, Relay Modules, Sensor Integration

Languages
Java, JavaScript, Python, C/C++, SQL, Bash

Certifications

- Certified Ethical Hacker (CEH) — EC-Council

Achievements

Second Prize, *ICTIH'26 International Conference*
Multiroom ESP32 Intercommunication Research Paper

CTF Competition Winner, *Red Team Hackers Academy*
Multiple wins at Red Team Hackers Academy

TryHackMe Top 4%
Ranked globally for ethical hacking and penetration testing